

RS004 Week 2 — English Worksheet

Topic: Gradient Background (Apply) · **Course:** Platformer Game · **Time:** about 45 minutes

This week your flat sky becomes a **gradient** — a smooth blend from one colour at the top to another at the bottom. You will gather all your colours into one **palette** and draw the sky one line at a time.

Keep these words handy: **gradient**, **palette**, **blend (interpolate)**, **draw.line**, **frame**.

1 · Predict 🧠

Read each snippet. Before you run it, write what you think will happen.

```
for y in range(HEIGHT):
    pygame.draw.line(screen, (100, 180, 255), (0, y), (WIDTH, y))
```

How many lines get drawn? What do they cover together?


```
t = y / HEIGHT
r = int(SKY_TOP[0] * (1 - t) + SKY_BOTTOM[0] * t)
```

At the very top (**y = 0**), what is **t** ? Which colour wins — top or bottom?


```
SKY_TOP = (100, 180, 255)
SKY_BOTTOM = (200, 230, 255)
```

These sit at the top of the file with the other colours. Why keep all colours in one place?

2 · Blend Math

The blend formula is `int(A * (1 - t) + B * t)`. Fill in the table. `A = 100`, `B = 200`.

`t = 0.0 → result = ____`

`t = 0.5 → result = ____`

`t = 1.0 → result = ____`

In your own words: when `t` goes from 0 to 1, the colour moves from __ to __.

3 · Spot the Bug 🐛

Each snippet is broken. Rewrite it so it works, then explain why the original was wrong.

Bug A — The sky should blend from top to bottom, but the whole screen comes out one flat colour.

```
for y in range(HEIGHT):
    t = 0
    r = int(SKY_TOP[0] * (1 - t) + SKY_BOTTOM[0] * t)
    g = int(SKY_TOP[1] * (1 - t) + SKY_BOTTOM[1] * t)
    b = int(SKY_TOP[2] * (1 - t) + SKY_BOTTOM[2] * t)
    pygame.draw.line(screen, (r, g, b), (0, y), (WIDTH, y))
```

Hint: `t` should *change* as `y` goes down. It is stuck at one value.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The gradient should fill the screen, but only the very top line is coloured.

```
y = 0
t = y / HEIGHT
r = int(SKY_TOP[0] * (1 - t) + SKY_BOTTOM[0] * t)
pygame.draw.line(screen, (r, SKY_TOP[1], SKY_TOP[2]), (0, y), (WIDTH, y))
```

Hint: one line is drawn. How do we draw *every* row?

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Modify the Sky

Start from your working `draw_gradient_background()` function. Make small changes and write down what each does.

1. Swap `SKY_TOP` and `SKY_BOTTOM`. What happens to the direction of the blend?

2. Pick two brand-new colours (a sunrise: warm at the bottom, dark at the top). Write the RGB values you chose.

3. **Stretch:** make a *three*-colour gradient (top → middle → bottom).

Hint: split the screen in half. Blend top→middle for the upper rows, then middle→bottom for the lower rows.

5 · Build & Show 📷

Build **your own** gradient background and run it on top of last week's window.

When it works, send a **photo or video**, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I chose my top and bottom colours: ...

The `for y in range(HEIGHT)` loop draws the sky by ...

The **t** value controls ...

I kept my colours in a palette so that ...

One tricky moment was ...

If I had more time, I would ...

6 · Record Your Walkthrough

Take a video while your gradient sky runs. Teach it to someone new. Try to use these words: **gradient, palette, blend, line, loop**.

1. Show your gradient sky.
2. Point at your two palette colours and say which is top, which is bottom.
3. Read the blend line out loud and explain what  does.
4. Show one colour you changed and the before/after.

Write what you will say in your video. Plan it here first.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.